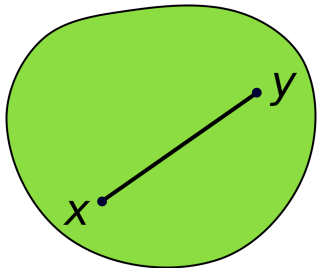


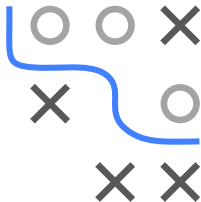
# Optimization in Machine Learning

## Foundations Convexity



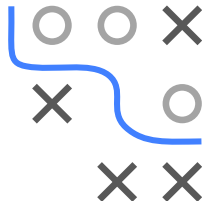
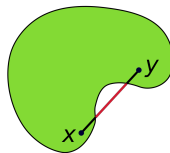
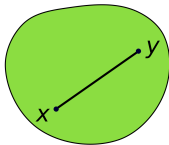
### Learning goals

- Convex sets
- Convex functions
- Strongly convex functions



# CONVEX SETS

- Set  $\mathcal{S} \subseteq \mathbb{R}^d$  is convex, if  $\forall \mathbf{x}, \mathbf{y} \in \mathcal{S}$  and  $\forall t \in [0, 1]$ :  
 $\mathbf{x} + t(\mathbf{y} - \mathbf{x}) \in \mathcal{S}$
- Intuition: Line between any  $\mathbf{x}, \mathbf{y} \in \mathcal{S}$  lies entirely in  $\mathcal{S}$
- Left: convex set; Right: not convex. (Source: Wikipedia)

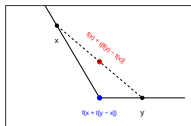
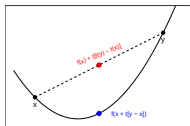


# CONVEX FUNCTIONS

- Let  $f : \mathcal{S} \rightarrow \mathbb{R}$ ,  $\mathcal{S}$  convex
- $f$  is convex if  $\forall \mathbf{x}, \mathbf{y} \in \mathcal{S}$  and  $\forall t \in [0, 1]$ :

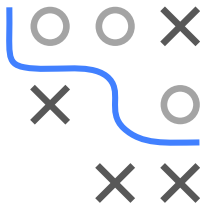
$$f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) \leq f(\mathbf{x}) + t(f(\mathbf{y}) - f(\mathbf{x}))$$

- Intuition: Connecting line for any  $\mathbf{x}, \mathbf{y} \in \mathcal{S}$  above function  $f$



Left: Strictly convex function; Right: Convex, but not strictly

- Strictly convex if  $<$  instead of  $\leq$
- Concave (strictly) if inequality holds with  $\geq$  ( $>$ )
- NB:  $f$  (strictly) concave  $\Leftrightarrow -f$  (strictly) convex

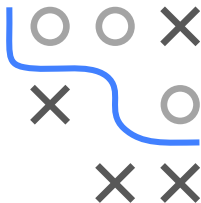
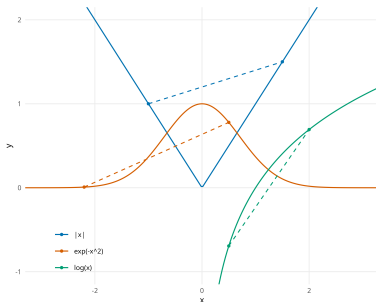


# SOME EXAMPLES

- Convex:  $f(x) = |x|$

$$\begin{aligned}f(x + t(y - x)) &= |x + t(y - x)| = |(1 - t)x + t \cdot y| \\&\leq |(1 - t)x| + |t \cdot y| = (1 - t)|x| + t|y| \\&= |x| + t \cdot (|y| - |x|) = f(x) + t \cdot (f(y) - f(x))\end{aligned}$$

- Concave:  $f(x) = \log(x)$
- Neither:  $f(x) = \exp(-x^2)$  (but log-concave)



# OPERATIONS PRESERVING CONVEXITY

- Nonnegatively weighted summation:

For  $w_1, \dots, w_n \geq 0$  and convex  $f_1, \dots, f_n$ :

$w_1 f_1 + \dots + w_n f_n$  is convex

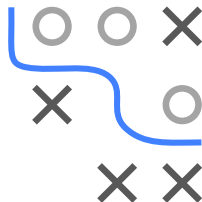
So: Sum of convex functions is also convex

- Composition:  $g$  convex,  $f$  linear:  $h = g \circ f$  is also convex:

$$\begin{aligned} h(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) &= g(f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))) \\ &= g(f(\mathbf{x}) + t(f(\mathbf{y}) - f(\mathbf{x}))) \\ &\leq g(f(\mathbf{x})) + t(g(f(\mathbf{y})) - g(f(\mathbf{x}))) \\ &= h(\mathbf{x}) + t(h(\mathbf{y}) - h(\mathbf{x})) \end{aligned}$$

- Element-wise maximization:  $f_1, \dots, f_n$  convex functions:

$g(\mathbf{x}) = \max \{f_1(\mathbf{x}), \dots, f_n(\mathbf{x})\}$  is also convex

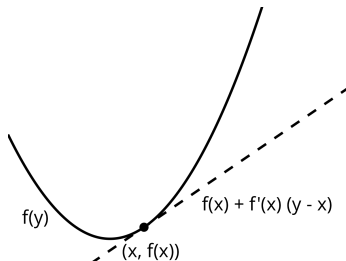


# FIRST ORDER CONDITION

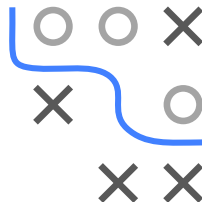
- For differentiable  $f$ , useful characterisation via gradient
- $f$  convex

$\iff$

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}) \text{ for all } \mathbf{x} \neq \mathbf{y} \in \mathcal{S}$$

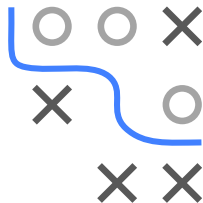


- Strictly convex if  $>$  instead of  $\geq$



## SECOND ORDER CONDITION

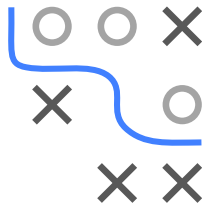
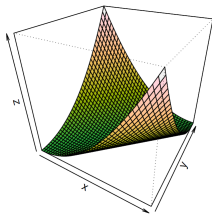
- For  $f \in \mathcal{C}^2$ : can characterize convexity via Hessian
- $f$  convex  $\iff H(\mathbf{x})$  psd  $\forall \mathbf{x} \in \mathcal{S}$
- $f$  strictly convex if  $H(\mathbf{x})$  pd  $\forall \mathbf{x} \in \mathcal{S}$
- To check global convexity, either verify the direct definition of psd by showing that  $\mathbf{v}^T H(\mathbf{x}) \mathbf{v} \geq 0$  for all  $\mathbf{v} \in \mathbb{R}^d$  and all  $\mathbf{x}$ , or, equivalently, check that all eigenvalues  $\lambda_i$  of all  $H(\mathbf{x})$  satisfy  $\lambda_i \geq 0$  for all  $\mathbf{x} \in \mathcal{S}$



# SECOND ORDER CONDITION

- Example:

$$f(\mathbf{x}) = x_1^2 + x_2^2 - 2x_1x_2; \quad \nabla f(\mathbf{x}) = \begin{pmatrix} 2x_1 - 2x_2 \\ 2x_2 - 2x_1 \end{pmatrix}^T; \quad H(\mathbf{x}) = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$$



- $f$  is convex since  $H(\mathbf{x})$  is p.s.d. for all  $\mathbf{x} \in \mathcal{S}$ :

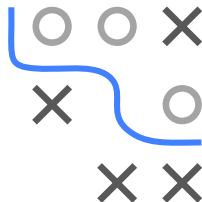
$$\begin{aligned} \mathbf{v}^T \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} \mathbf{v} &= 2v_1^2 - 2v_1v_2 - 2v_1v_2 + 2v_2^2 \\ &= 2v_1^2 - 4v_1v_2 + 2v_2^2 = 2(v_1 - v_2)^2 \geq 0 \end{aligned}$$

# FIRST-ORDER OPTIMALITY (CONVEX CASE)

- For convex differentiable  $f$ : vanishing gradient is sufficient for a global min
- **Theorem.** Let  $f$  be continuously differentiable and convex.  
If  $\nabla f(\mathbf{x}^*) = 0$ , then  $\mathbf{x}^*$  is a global minimizer.
- **Proof.** Set  $\mathbf{x} = \mathbf{x}^*$  in the first-order characterization:

$$f(\mathbf{y}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)(\mathbf{y} - \mathbf{x}^*) = f(\mathbf{x}^*) \quad \forall \mathbf{y} \in \mathcal{S}$$

- Hence for convex  $f$ : first-order necessary  $\Leftrightarrow$  first-order sufficient



# STRONG CONVEXITY

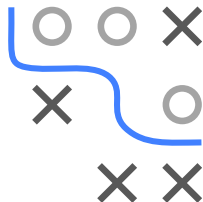
- $f$  is **strongly convex** with modulus  $m > 0$  if  $\forall \mathbf{x}, \mathbf{y} \in \mathcal{S}, t \in [0, 1]$ :

$$f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) \leq f(\mathbf{x}) + t(f(\mathbf{y}) - f(\mathbf{x})) - \frac{1}{2} m t(1 - t) \|\mathbf{x} - \mathbf{y}\|^2$$

- Stronger than convex: extra negative term  $-\frac{1}{2} m t(1 - t) \|\mathbf{x} - \mathbf{y}\|^2$  on the right
- For differentiable  $f$ , equivalently:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}) + \frac{m}{2} \|\mathbf{y} - \mathbf{x}\|^2$$

- **Interpretation:**  $f$  is lower-bounded by a quadratic touching  $f$  at  $\mathbf{x}$  (complements the upper bound from  $L$ -smoothness; cf. descent lemma)



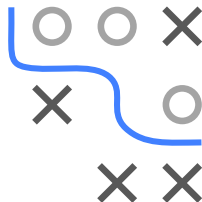
# STRONG CONVEXITY: HESSIAN AND UNIQUENESS

- For  $f \in \mathcal{C}^2$ :  $f$  is  $m$ -strongly convex  $\iff H(\mathbf{x}) \succeq m\mathbf{I}_d \ \forall \mathbf{x} \in \mathcal{S}$
- Equivalently, all eigenvalues of  $H(\mathbf{x})$  are bounded below by  $m > 0$
- Combined with  $L$ -smooth  $\iff H(\mathbf{x}) \preceq L\mathbf{I}_d$ :

$$m\mathbf{I}_d \preceq H(\mathbf{x}) \preceq L\mathbf{I}_d$$

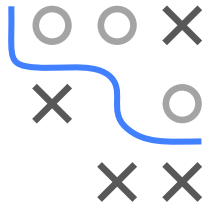
- Ratio  $\kappa = L/m$  is the **condition number** of  $f$  (later: governs convergence rate)
- **Theorem.** For strongly convex  $f$ , if  $\nabla f(\mathbf{x}^*) = 0$ , then  $\mathbf{x}^*$  is the **unique** global minimizer.
- **Proof.** From the gradient form with  $\mathbf{x} = \mathbf{x}^*$  and  $\nabla f(\mathbf{x}^*) = 0$ :

$$f(\mathbf{y}) \geq f(\mathbf{x}^*) + \frac{m}{2}\|\mathbf{y} - \mathbf{x}^*\|^2 > f(\mathbf{x}^*) \quad \forall \mathbf{y} \neq \mathbf{x}^*$$



# CONVEX FUNCTIONS IN OPTIMIZATION

- Will see later:
- For a convex function, every local optimum is also a global one  
⇒ No need for involved global optimizers, local ones are enough
- A strictly convex function has at most one optimal point
- “... in fact, the great watershed in optimization isn’t between linearity and nonlinearity, but convexity and nonconvexity.”  
– R. Tyrrell Rockafellar. *SIAM Review*, 1993



SIAM REVIEW  
Vol. 35, No. 2, pp. 183–238, June 1993

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## LAGRANGE MULTIPLIERS AND OPTIMALITY\*

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**Abstract.** Lagrange multipliers used to be viewed as auxiliary variables introduced in a problem of constrained minimization in order to write first-order optimality conditions formally as a system of equations. Modern applications, with their emphasis on numerical methods and more complicated side conditions than equations, have demanded deeper understanding of the concept and how it fits into a larger theoretical picture.

A major line of research has been the nonsmooth geometry of one-sided tangent and normal vectors to the set of points satisfying the given constraints. Another has been the game-theoretic role of multiplier vectors as solutions to a dual problem. Interpretations as generalized derivatives of the optimal value with respect to problem parameters have also been explored. Lagrange multipliers are now being seen as arising from a general rule for the subdifferentiation of a nonsmooth objective function which allows black-and-white constraints to be replaced by penalty expressions. This paper traces such themes in the current theory of Lagrange multipliers, providing along the way a free-standing exposition of basic nonsmooth analysis as motivated by and applied to this subject.

**Key words.** Lagrange multipliers, optimization, saddle points, dual problems, augmented Lagrangian, constraint qualifications, normal cones, subgradients, nonsmooth analysis

**AMS subject classifications.** 49K99, 58C20, 90C99, 49M29